Algorithms and Data Structures: Quiz 2

Name: _____ Class: _____

Part 1. Mark exactly one option per question with an X.

1. Which structure removes the most recently added item first?

☐ A) Queue ☐ B) Stack ☐ C) Heap ☐ D) Tree ☐ E) Graph

2. Worst-case time of linear search on n items:

☐ A) O(1) ☐ B) O(log n) ☐ C) O(n) ☐ D) O(n log n) ☐ E) $O(n^2)$

3. Which of these sorting algorithms is stable?

☐ A) Quick ☐ B) Heap ☐ C) Merge ☐ D) Selection ☐ E) Shell

4. Average time of a lookup in a hash table:

☐ A) O(1) ☐ B) O(log n) ☐ C) O(n) ☐ D) O(n log n) ☐ E) $O(n^2)$

Write clearly; answers that cannot be read are not graded.
Part 2 is on the next page.

Part 2: explain your answers

5. What is the time complexity of binary search, and why? (3 points)

6. A recursive function has no base case. What happens when it runs, and why? (2 points)

7. What is the difference between a stack and a queue? Give one real-world example of each. (2 points)